MOHAMED BASSAM HASSAN ABDELMEGEED

AI ENGINEER | MACHINE LEARNING | COMPUTER VISION | NLP | AGENTIC AI

Dubai, UAE | UAE Golden Visa Holder | Available for Immediate Joining

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PROFESSIONAL SUMMARY

AI Engineer and Full-Stack Developer with hands-on experience building and deploying end-to-end AI systems across agentic AI, machine learning, deep learning, computer vision, natural language processing, retrieval-augmented generation, and SaaS platforms. I have skills in Python, PyTorch, TensorFlow, FastAPI, Flask, LangGraph, PostgreSQL/pgvector, Docker, and Azure. Built production-style multi-agent, multimodal compliance, financial forecasting, and real-time safety monitoring systems with secure APIs, automated testing, and cloud-ready architecture.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL

Machine Learning and Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, LSTM, BERT, model evaluation, feature engineering, time-series forecasting

Agentic AI and NLP: LangGraph, multi-agent systems, RAG, hybrid retrieval, vector databases, OpenAI API, Anthropic API, Hugging Face Transformers, MCP, prompt engineering

Computer Vision: YOLO11, OpenCV, object detection, PPE detection, real-time inference, CUDA

Backend and SaaS: FastAPI, Flask, REST APIs, WebSockets, Pydantic, APScheduler, Clerk, Stripe, Next.js, React, Tailwind CSS

Databases, Cloud and DevOps: PostgreSQL, pgvector, SQLite, BM25, Reciprocal Rank Fusion, Docker, Azure Service Bus, Azure Key Vault, Git, GitHub, Pytest, n8n

PROFESSIONAL EXPERIENCE

Machine Learning Intern | CodeAlpha | Remote | 2024

- Developed, tested, and evaluated machine learning models using support vector machines, neural networks, TensorFlow, Keras, and Scikit-learn.
- Built practical object recognition and data classification workflows using OpenCV, NumPy, Pandas, and Matplotlib.
- Evaluated classification performance using ROC curves, confusion matrices, precision, recall, and related model metrics.

SELECTED AI PROJECTS

Dark Agent - Governed Multi-Agent AI Mission Control Platform

Technologies: Python, FastAPI, LangGraph, Next.js, TypeScript, PostgreSQL, pgvector, BM25, RRF, WebSockets, Clerk, Stripe, Docker, Azure, MCP, OpenAI, Anthropic

- Architected a governed four-agent LangGraph workflow with a Planner, parallel specialists, a Synthesizer, and an adversarial Red Team validator, including human plan approval and real-time WebSocket streaming.
- Implemented tenant-isolated hybrid RAG using pgvector semantic search, BM25 keyword ranking, Reciprocal Rank Fusion, episodic memory distillation, and PII sanitization.
- Secured the SaaS control plane with authentication, idempotent subscription webhooks, model routing, rate limits, deterministic audit logs, and zero-retention Ghost Mode.
- Built Docker and Azure production foundations with Service Bus workers, Key Vault-backed connectors, durable queues, provider fallback, 309 backend tests, and a verified Next.js production build.

Contract Verifier AI - Multimodal Compliance and Verification Engine

Technologies: FastAPI, Twilio WhatsApp Webhooks, Google Gemini Vision, ElevenLabs TTS, Next.js, asynchronous processing

- Built a document verification pipeline that extracts structured legal terms from mobile contract images and detects potentially harmful discrepancies.
- Engineered a sub-10-second image-to-audio workflow that delivers localized voice alerts through WhatsApp.
- Improved resilience with asynchronous queues, sensitive-data log sanitization, health endpoints, and a real-time operations dashboard.

AI Stock Price Prediction - Market Analysis and Paper Trading Platform

Technologies: Python, Flask, TensorFlow, Scikit-learn, XGBoost, LightGBM, Alpaca API, Chart.js, time-series forecasting

- Developed a multi-model ensemble using LSTM, XGBoost, LightGBM, and ExtraTrees to forecast market trends and generate explainable trade setups.
- Integrated Alpaca Paper Trading for simulated order execution, portfolio synchronization, and real-time profit-and-loss tracking.
- Implemented walk-forward validation, historical backtesting, market-regime analysis, technical indicators, and confidence-based prediction explanations.

AI Industrial Safety Monitoring System

Technologies: Python, YOLO11, OpenCV, PyTorch, CUDA, n8n

- Developed a real-time PPE monitoring system for hardhat and no-hardhat detection in safe and danger zones.
- Implemented spatial violation rules, dual-model validation, incident logging, and automated Gmail alerts with evidence images through n8n.
- Optimized inference performance using CUDA acceleration and reduced false alerts through validation logic and alert cooldown controls.

ADDITIONAL PROJECTS

AI Study Assistant

Built an NLP summarization application using Hugging Face BART, Flask APIs, PDF upload processing, and document chunking for long files.

AI Writer SaaS

Developed and deployed a Flask and SQLite application with secure authentication, session management, generation APIs, and persistent user history on PythonAnywhere.

EDUCATION

Bachelor of Computer Science (Artificial Intelligence)

Arab Open University | Graduated July 2025

Dual Degree Program verified by The Open University, United Kingdom.